

1. Describe the different types of test data generators.

- **Random Test Data Generators**
 - Randomly **generate bit streams and convert into input type needed.**
 - Can be used to **generate input for any type** of program.
 - Disadvantage: **does not generate quality data** (low coverage) – solely based on probability
- **Goal Oriented Test Data Generators**
 - Using **CFG to generate input for any path specified** instead of just the usual way of generating input from the entry to the exit of the code.
 - To **find paths effectively and automatically**
 - Disadvantage: May generate infeasible paths (**too many paths**).
- **Pathwise Test Data Generators**
 - Similar to the Goal Oriented test data generator, but **using a specific path** (assigned specific path to follow) instead of giving a choice among many paths.
 - Disadvantage : Better but **harder to generate the needed test data** due to greater path knowledge and prediction of coverage.
- **Intelligent Test Data Generators**
 - Depend on sophisticated **analysis of the code to be tested to search for test data.**
 - Disadvantage: **complex** and need insights in order to anticipate the different problems

2. When selecting test data, there are some questions to be asked for the test data requirements. What are them?

- Use **real data**?
- Generate data at **random**?
- Select data from a **fixed pool**?
- Selection based on **data characteristics**

3. Explain the importance of test environment readiness review.
 - It ensures the test setup is complete, stable, and mirrors the production environment. It helps produce accurate and reliable test results, prevents delays caused by missing tools or data and reduces the risk of invalid outcomes.
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4. Analyse the challenges in using production data.
 - Not so good at a unit or acceptance testing level (large amount)
 - Difficult to express intent (test objective)
 - Data ends up completely **externalized** from the test
5. Discuss the guidelines for test environment readiness.
 - Stakeholder requirements collection and addressing:
 - Multiple teams will be working on different projects, testing and **proper management** of these stakeholders should happen.
 - Centralized way of managing environment and test data requirements:
 - **Multiple teams** would be working in **multiple projects** and at times, data requirements will be overlapped.
 - Test environment management strategy:
 - A clear-cut and effective way of **managing and providing guidance to teams** should be defined.
 - As part of **overall monitoring**, periodic utilization data should be analyzed so that if a particular environment is not utilized effectively.
 - Multiple types/round/cycle of testing:
 - Definitely affect the testing environment. Therefore, the need is to **identify which data can be shared, which data is exclusive in nature.**
6. Analyse the effects of inappropriate setup of test environment.
 - Inaccurate Test Results and False Confidence
 - Increase Cost
 - Reduced Test Efficiency and Effectiveness
 - Security Vulnerabilities